

Fire Brigade

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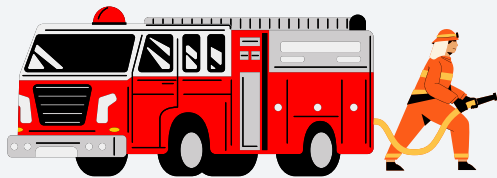
School of Engineering



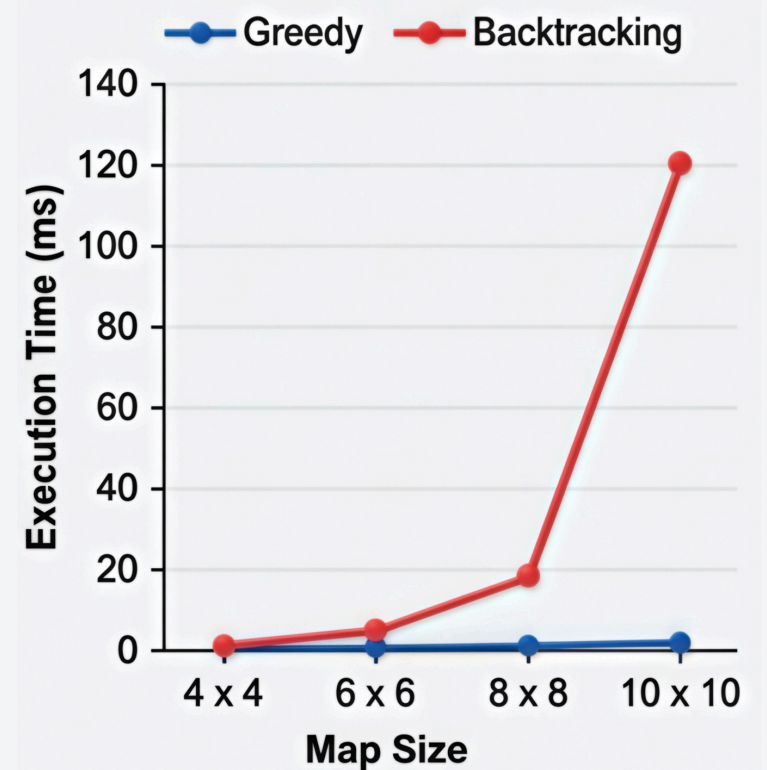
Introduction

Fire Brigade is a turn-based tactical strategy game developed on an **8 x 8** grid. The system simulates a dynamic environment where fire spreads while the player attempts to mitigate damage and rescue civilians using limited resources.

Victory is achieved by extinguishing all fire sources and successfully evacuating all civilians. Defeat occurs if fire consumes more than 80% of the map or if more than 50% of the initial civilian population is lost.



Results



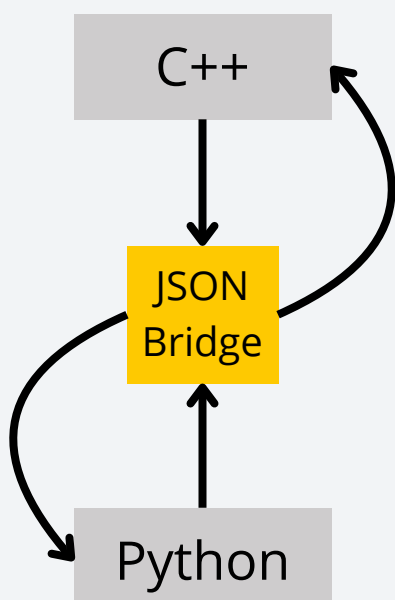
The hybrid architecture successfully optimizes performance.

While C++ handles spatial structures (**BST and Linked Lists**) with minimal overhead, Python's decision logic yields contrasting results, the **Greedy** approach provides instant reactive responses ideal for rapid fire spread, whereas **Backtracking** guarantees optimal civilian rescue paths at the expense of higher computational complexity in critical scenarios.

Goal

Develop a high-performance tactical simulation using a hybrid C++/Python architecture, where custom-built data structures (**BST and Linked Lists**) and optimization algorithms (**Greedy and Backtracking**) drive the real-time decision-making for wildfire containment and civilian rescue.

System Architecture



The system is divided into two main components:

- C++ (**Game Engine**): Managing the Game State, Fire spread, BST of risk areas, Linked list of units.
- Python (**Decision Logic**): Greedy Algorithm, Backtracking algorithm.

Communication: JSON Bridge (input.json / state.json)

Conclusion

A high-performance simulation was achieved using a hybrid C++/Python architecture. The JSON Bridge enabled real-time execution, while custom **data structures** optimized map management. The implemented **Greedy** and **Backtracking** algorithms successfully resolve dynamic wildfire containment and civilian rescue.



For more information
Click on the QR code